

15 Oscillatory Motion

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$$F = -kx$$

$$F = m a_x$$

$$-kx = ma$$

$$a = -\frac{k}{m} x$$

$$\frac{d^2 x}{dt^2} = -\frac{k}{m} x$$

$$\frac{k}{m} \equiv \omega^2$$

$$\frac{d^2 x}{dt^2} = -\omega^2 x$$

$$x(t) = A \cos(\omega t + \varphi)$$

$$\frac{dx}{dt} = -\omega A \sin(\omega t + \varphi)$$

$$\frac{d^2 x}{dt^2} = -\omega^2 A \cos(\omega t + \varphi)$$

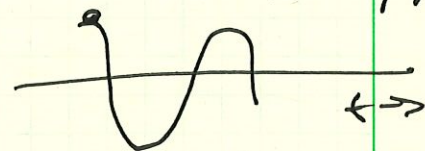
✓

$\omega t + \varphi$ is the phase

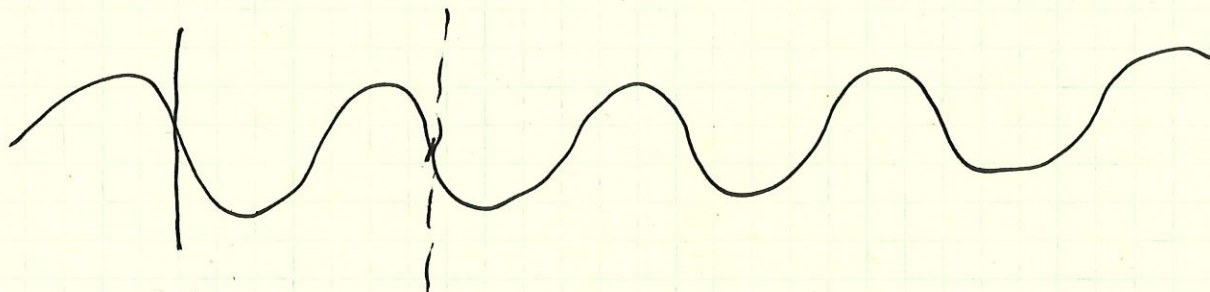
φ is phase constant

A amplitude

If particle at $x=A$ at $t=0$



then $\phi=0$



$t=0$

T

period

$\uparrow$

$\omega t + \phi$

$\uparrow$

$\omega(t+T) + \phi$

$$2\pi = \Delta \text{phase} [\omega(t+T) + \phi] - [\omega t + \phi] = \omega T$$

$$T = \frac{2\pi}{\omega}$$

$$T \equiv \frac{1}{f} \quad \text{frequency}$$

$$\omega = \frac{2\pi}{T}$$

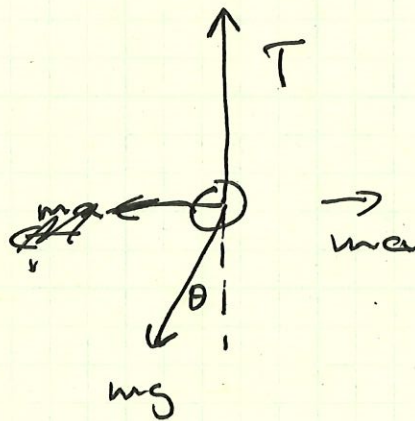
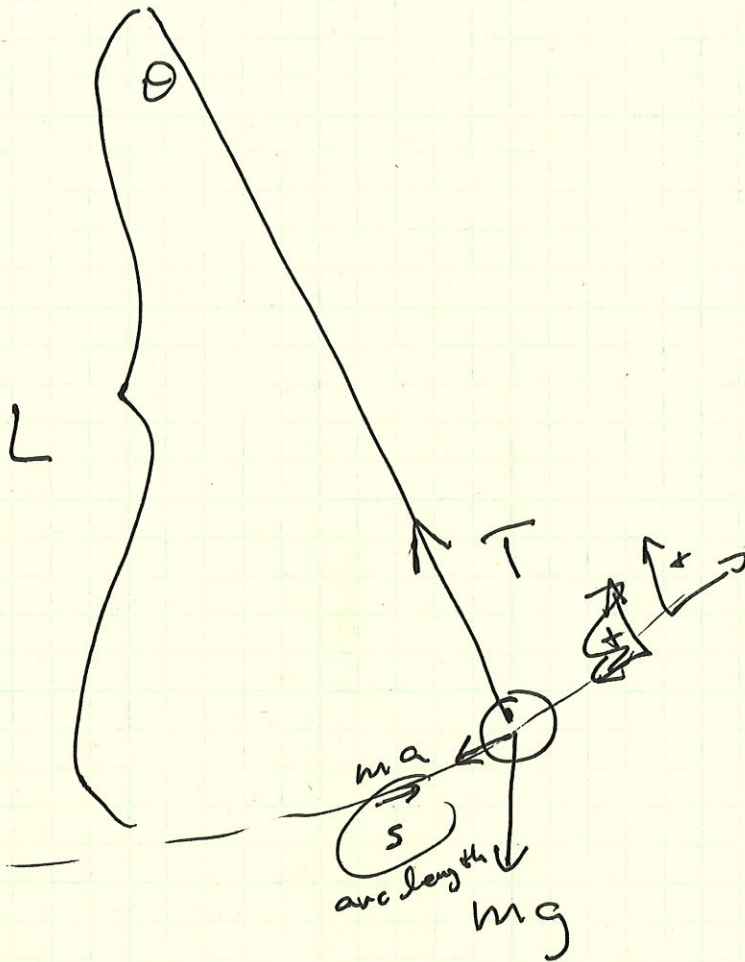
$$f = \frac{\omega}{2\pi}$$

$$\boxed{f = \frac{1}{T}} = \frac{\omega}{2\pi}$$

$$f = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

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$$F_t = m a_t \quad \text{tangential}$$

$$-mg \sin \theta = m \frac{d^2 s}{dt^2}$$

$$s = L \theta$$

$$-g \sin \theta = L \frac{d^2 \theta}{dt^2}$$

$$\frac{d^2 \theta}{dt^2} = -\frac{g}{L} \sin \theta \approx -\frac{g}{L} \theta$$

$$\sin \theta \sim \theta \quad ??$$

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$$\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \dots$$

truncate at θ

so only for small θ

$$\frac{d^2 \theta}{dt^2} = -\frac{g}{L} \theta$$

$$\theta = \theta_{\max} \sin(\omega t + \phi)$$

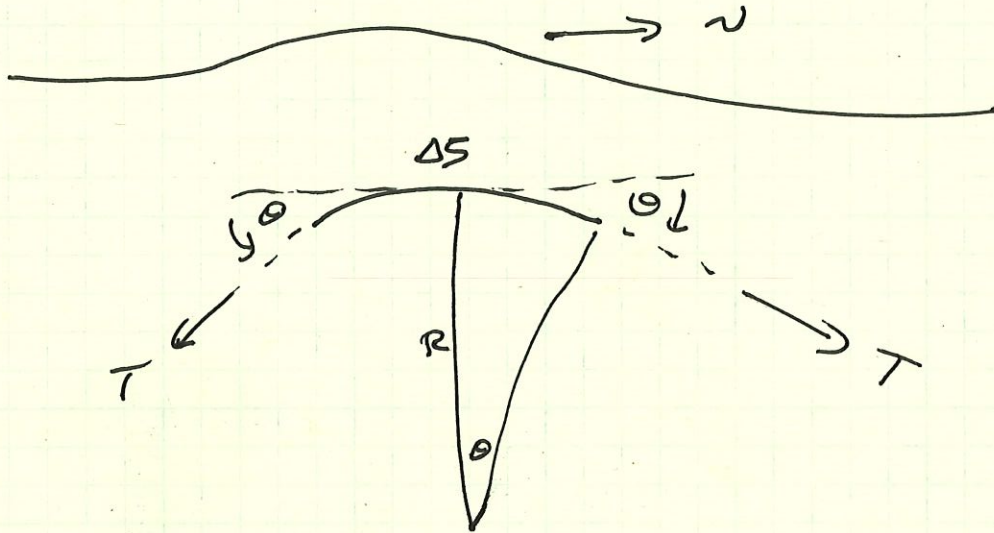
$$\omega = \sqrt{\frac{g}{L}}$$

$$T = 2\pi \sqrt{\frac{L}{g}}$$

for θ small

HW #3

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$$F_r = 2T \sin \theta = 2T \theta$$

$$m = \mu \Delta s = 2\mu R \theta$$

$$F_r = \frac{mv^2}{R}$$

$$2T \theta = 2\mu R \theta \frac{v^2}{R}$$

$$T = \mu v^2$$

$$v = \sqrt{\frac{T}{\mu}}$$

speed of wave in
stretched wire
or cord!

$$v = 331 \left(1 + \frac{T_c}{273} \right)$$

$$v \sim 343 \text{ m/s}$$

$$T = 20^\circ\text{C}$$

① observer moves

$$f' = \frac{v'}{\lambda} = \frac{v + v_o}{\lambda}$$

v sound

v_o observer

$$f' = \frac{v + v_o}{v} f$$

$$\lambda = \frac{v}{f}$$

observer moves

toward source

$$f' = \frac{v - v_o}{v} f$$

observer moves

away from source

② What if source moves?

λ' gets smaller

as source approaches
at v_s

$$\lambda' = \lambda - \frac{v_s}{f}$$

in λ source moves

$$v_s T = \frac{v_s}{f}$$

$$f' \text{ heard} \quad f' = \frac{v}{\lambda'} = \frac{v}{\lambda - \frac{v_s}{f}} = \frac{v}{\frac{v}{f} - \frac{v_s}{f}}$$

$$f' = \frac{v}{v - v_s} f$$

source toward
observer

$$f' = \frac{v}{v + v_s} f$$

source away
from observer

together

$$f' = \frac{v + v_o}{v - v_s} f$$

+ for v_o or v_s
toward obs

- for v_o or v_s
away from obs